DELOITTE CYBER GURUKUL MCQ SYLLABUS ANALYSIS

Classification of Regularly Asked vs. Rarely Injected Core Technical MCQ Topics (HirePro)

SECTION 1: COMPUTER NETWORKS

REGULARLY ASKED TOPICS

1. OSI Layer Data Units and Header Analysis

Focus: Determining what data forms look like at various stages of transmission. You must immediately identify that Layer 2 maps to *Frames*, Layer 3 maps to *Packets*, and Layer 4 maps to *Segments*.

- **Sample Question Prompt:** If a network packet passes through a router interface, which headers are rewritten? (Answer: Layer 2 MAC addresses are altered; Layer 3 IP headers remain unchanged unless NAT is applied).

2. Standard Port Mapping & Secure Discrepancies

Focus: Direct comparison between encrypted secure operational channels and legacy cleartext endpoints.

- **Key Mappings:** SSH/SFTP (22) vs Telnet (23) and FTP (21). HTTP (80) vs HTTPS (443). DNS (53), SMTP (25), RDP (3389).

3. TCP 3-Way Handshake & Control Flags

Focus: Tracing the sequential bits flipped inside the TCP header during state transitions.

- **Sequence:** SYN → SYN-ACK → ACK. Questions frequently present a scenario where an unexpected connection drop happens and ask which flag is thrown (Answer: RST - Reset).

RARELY INJECTED TOPICS

4. Subnet Mask Calculations and CIDR Notation

Focus: Deep network partitioning math. Finding network addresses or total hosts given boundaries like `/27` or `/29`.

5. BGP and OSPF Routing States

Focus: Distance-vector vs Link-state internal pathing parameters. Rarely asked but appears if HirePro serves a high-tier routing profile.

SECTION 2: INFORMATION SECURITY & CRYPTOGRAPHY

REGULARLY ASKED TOPICS

1. Asymmetric Public-Key Directionality

Focus: Knowing precisely which key executes confidential scrambling versus structural authentication signatures.

Objective	Encryption Phase (Sender Side)	Decryption Phase (Receiver Side)
Confidentiality (Privacy)	Recipient's Public Key	Recipient's Private Key
Authentication (Digital Signature)	Sender's Private Key	Sender's Public Key

2. Hash Function Security Pillars

Focus: Properties of MD5, SHA-1, and SHA-256. Differentiating between *Collision Resistance* (finding two arbitrary texts matching digests) and *Pre-image Resistance* (reversing a hash digest back to plaintext).

RARELY INJECTED TOPICS

3. Symmetric Key Count Scaling Formula

Focus: Calculating how many unique symmetric keys are required for secure peer communication within a group of n individuals. Formula: `N * (N - 1) / 2`.

4. Block Cipher Modes of Operation

Focus: Comparing Electronic Codebook (ECB) mode, which produces identical blocks under the same key, against Cipher Block Chaining (CBC) mode using Initialization Vectors (IVs).

SECTION 3: OPERATING SYSTEMS & SECURE CODING

REGULARLY ASKED TOPICS

1. Octal Linux File Permissions Transformation

Focus: Reading symbolic strings and executing rapid numeric assignments based on weights: Read (4), Write (2), Execute (1).

- **Example:** String `rwxr-x---` maps to Owner: 4+2+1=7, Group: 4+0+1=5, Others: 0. Hex/Octal code = `750`.

2. OWASP Top 10 Software Flaws Analysis

- **Buffer Overflow:** Unchecked memory bounds rewriting stack frames and corrupting execution return pointers.
- **SQL Injection (SQLi):** Inputting backend escape sequences to alter running database queries.
- **Cross-Site Scripting (XSS):** Feeding arbitrary JavaScript parameters into application windows to hijack user sessions.

RARELY INJECTED TOPICS

3. Static vs Dynamic Scope Resolution

Focus: Tracking pseudo-code variables when a function accesses an outer variable block. Static scope checks structural nesting layout; dynamic scope walks up the current running call stack.